Regional Adaptive Pathways Planning Project (RAPP)

Defend? Retreat? Accommodate? How do we plan for future regional settlement in a changing climate

Future of Place: Changing Cities, Towns And Regional Settlements

Presented by
Peter Heinmiller &
Christopher Marinopoulos

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Delivery Partners

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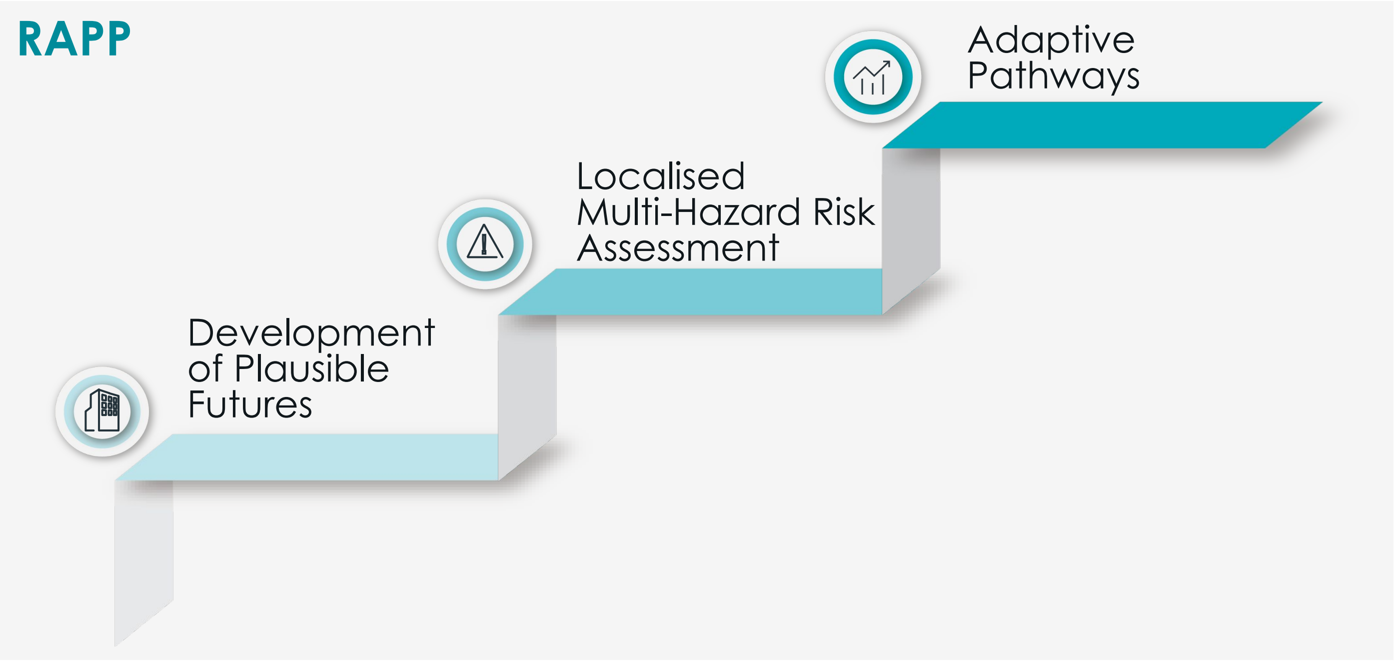
Strategic Planning through a Multi-Hazard and Resilient Lens: Accessible Practices to Achieve New Ways of Planning for the Future



Current Strategic Planning Practices

- Limited integration between Local Councils
- Minimal incorporation of natural disaster risk and hazards
- Shorter Term timeframes
- Limited incorporation of Sustainability measures (i.e., UN SDGs, etc.)

Incorporating Resilience in Strategic Planning

Connected Engagement through Planning Processes

Accessible, Inclusive, Usable

Source: Value Advisory Partners 2025

Evaluation framework for strategic planning through RAPP project

- Planning in context of the plausible futures (low growth and high growth)
- Support progress towards achieving the First Nations principles, community vision and relevant UN SDGs in the long term
- Understand benefits associated with enabling infrastructure and hazard mitigation

Source: United Nations Department of Economic and Social Affairs, Global Sustainable Development Goals, 2025

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Challenges For The Future Of Planning for Changing Cities, Towns And Regional Settlements

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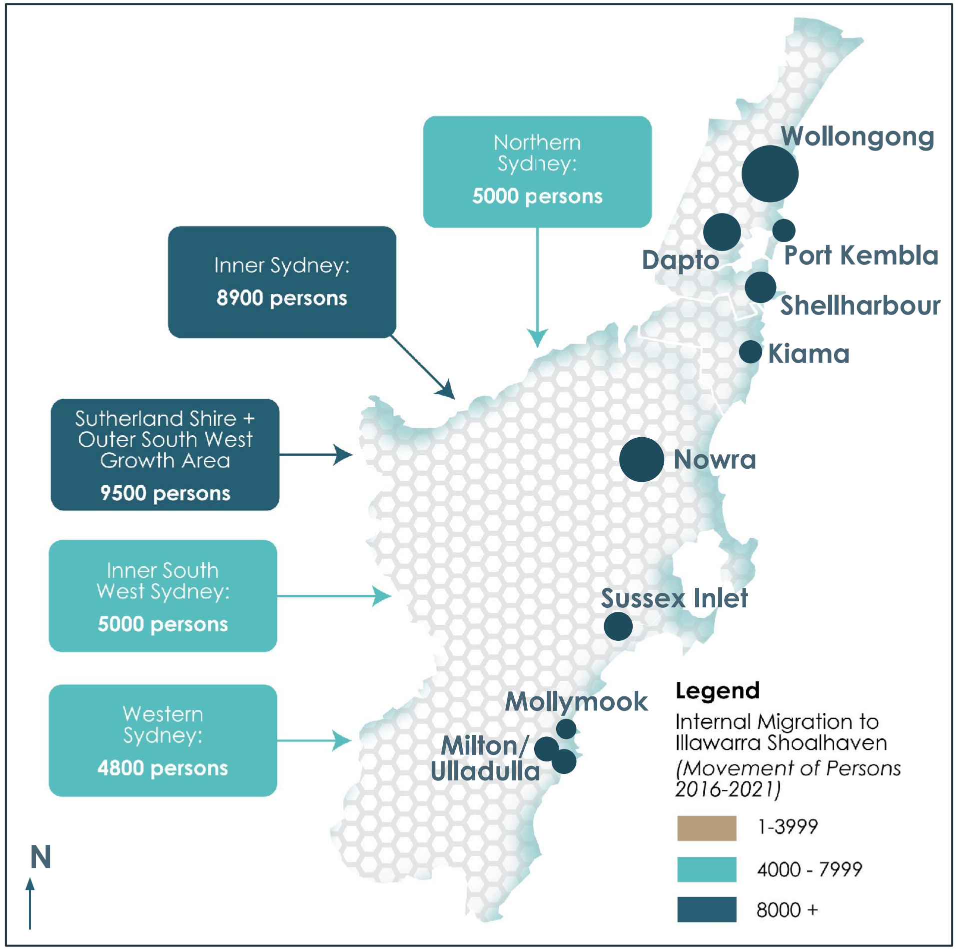
Accelerating Housing Growth including Internal Migration within Multiple Growth Areas of Illawarra Shoalhaven

Key Challenges

- **Where to accommodate** housing growth within the region
- Population moving to regions **placing stress** on existing areas
- Illawarra's connection to Sydney, Illawarra as a "**commuter suburban area**" due to housing affordability
- **Limited collaboration and shared understanding** between levels of government
- COVID and Post-COVID accelerating "**Sea change**" & "**Tree change**" from city to regional towns in growing numbers

5 Year Internal Migration to Illawarra Shoalhaven



Source: ABS Census Data, 2021; Value Advisory Partners 2025

Growth examples

Infill: Wollongong CBD

Greenfield: Dapto Growth Area

Expanding Townships: Mollymook

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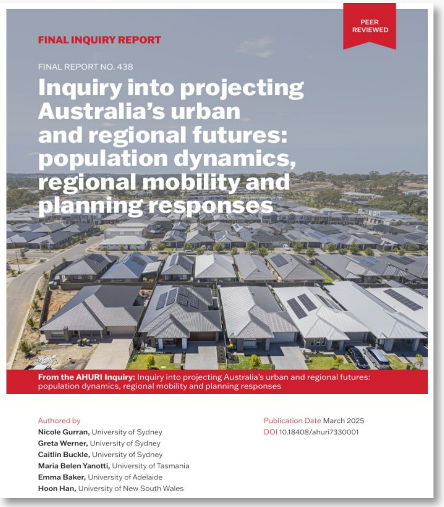
Planning for Increased Infrastructure and Amenity Needs for New Housing Growth and Existing Urban Areas

Need For Change

- **Place-based decisions** on how housing growth affects regions, "not just like the cities"
- Planning for housing growth can occur **at multiple scales** where policies could be as local as a neighbourhood precinct, or as broad as statewide policy
- Policies and strategies need to address both wider objectives about housing as well as **incorporating local issues**
- Identifying **Infrastructure** and Amenity to **support growth**
- Current high levels of demand for housing **limits current housing affordability**

"Contemporary drivers of regional mobility and housing market spillovers, and implications for infrastructure and settlement planning" (AHURI, 2025)



AHURI - Inquiry into projecting Australia's urban and regional futures: population dynamics, regional mobility and planning responses

Published March 2025

Planning for the future of place at multiple scales

State
- Multiple Urban Areas Planning Policies
- State Environment Planning Policies

Cities & Regions
- City-wide (Urban area) Planning Policies
- Regional Planning Policies

Local
- Local Strategic Planning Statement
- Local Infrastructure Contributions Plans

Precincts within LGAs
- State Significant Precincts
- Precinct Plans

Increased Frequency and Intensity of Multiple Hazards having an Overwhelming Effect on Cities, Towns and Regional Settlements

Key challenges

- The Illawarra Shoalhaven region has been directly hit with **catastrophic natural disasters within the last 5 years**
- This is through **events from multiple hazards,** with some specific areas within the region hit by multiple events in same year
- Regional Planning needs to consider hazards that are becoming **more frequent** and **higher in intensity** when planning for changes in land use and housing settlement
- Including multi-hazard assessments can lead to more **place-based housing strategies** for regions

Examples of recent hazards in the Illawarra Shoalhaven Region

Bushfire: Lake Conjola

Source: https://www.news.com.au/national/northern-territory/behind-the-lens-of-the-australian-bushfire-crisis/image-gallery/d5b4ccc2392d75e42444f0019d663a4d?page=17/

Riverine Flooding: Nowra

Source: https://www.southcoastregister.com.au/story/3954904/shoalhaven-declared-natural-disaster-zone-photos/

Storm Surge: Lake Illawarra

Source: https://www.ulladullatimes.com.au/story/6621722/shoalhaven-floods-evacuation-warning-issued-for-parts-of-lake-conjola/

Landslides: Wattamolla

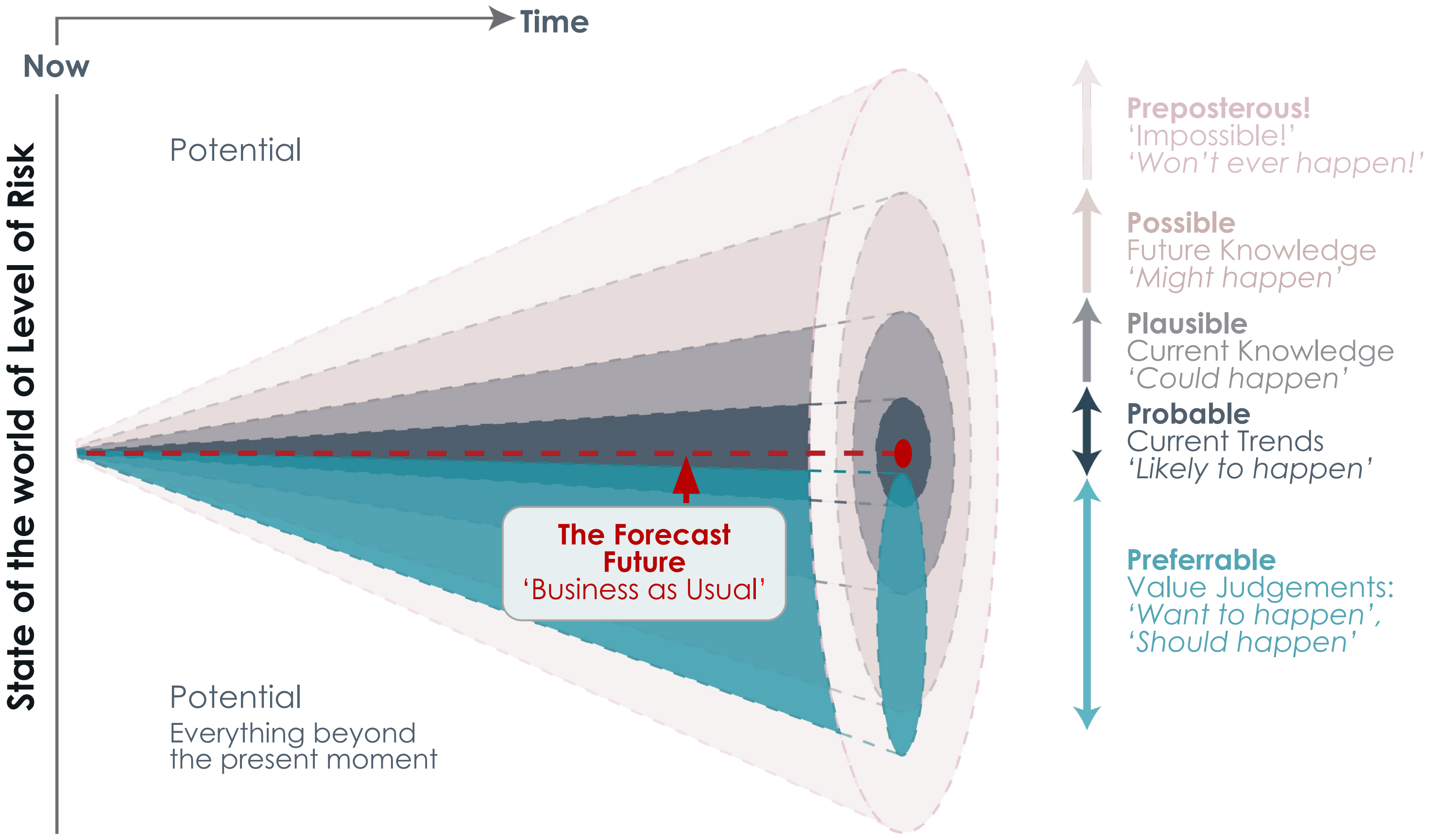
Source: https://www.southcoastregister.com.au/story/7721472/wattamolla-road-could-be-closed-for-up-to-three-months/

Need for New Processes to Deal with Increased Uncertainty in Long Term Planning to Future Proof Cities, Towns And Regional Settlements

Key Challenges

- Increased uncertainty comes from both **increased hazard** and **increased need for housing** in the regions into the future
- **Key changes:**
  - Population/housing demand
  - Enabling/supporting infrastructure
  - Climate and hazard
  - Socio-economic and technology
- **To achieve a preferable and resilient future, adaptation is required to deal with inherent uncertainty**

The Futures Cone – Illustrating the Spectrum of Potential Futures



Source: Department of Home Affairs, National Resilience Taskforce; Adapted from Voros, J. 2003. "A generic foresight process framework'." Foresight 5 (3):pp. 10-21;

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Innovation And Transformative Processes within the Regional Adaptive Pathways Planning Project (RAPP)

Vision and Objectives for RAPP

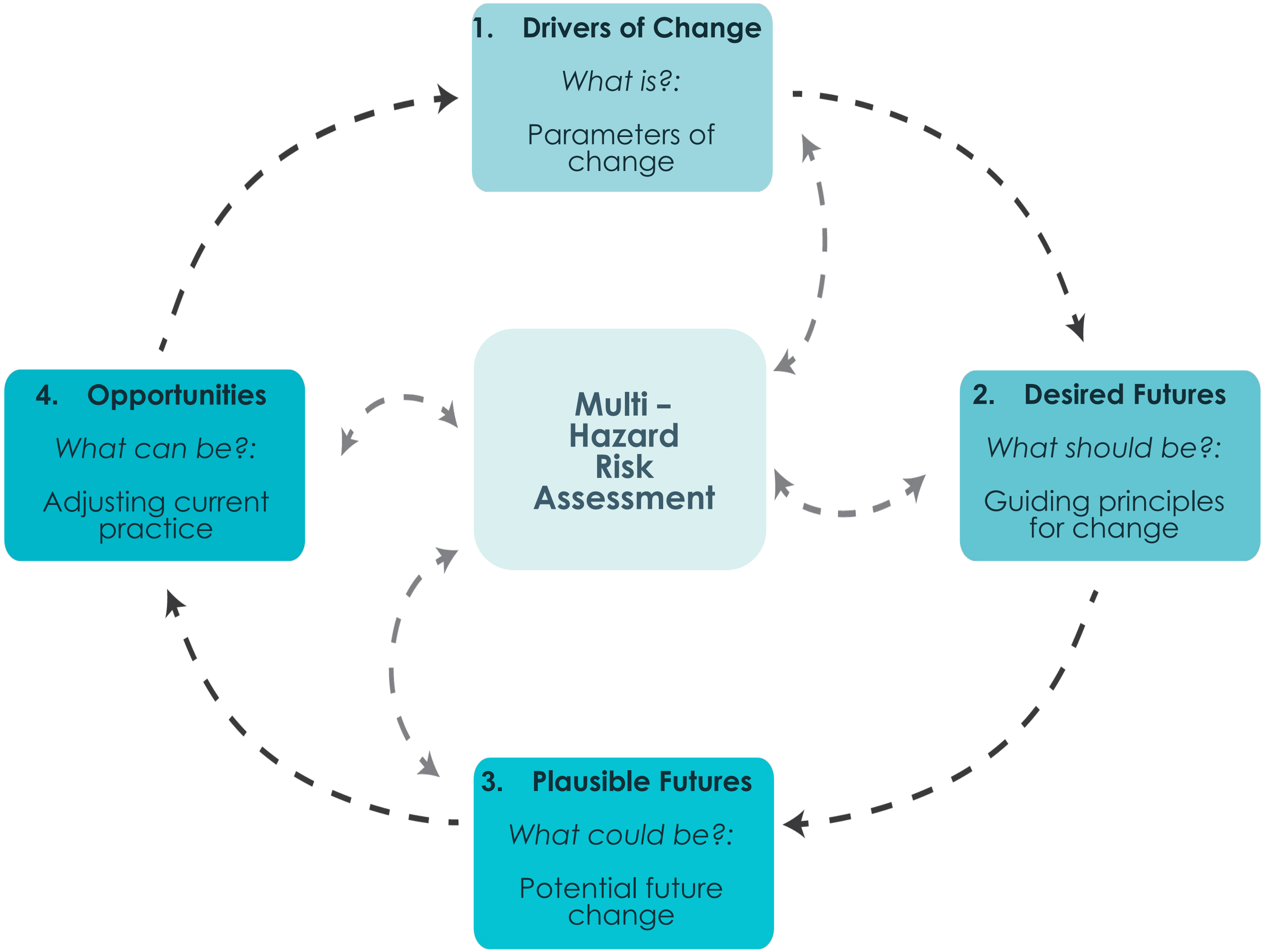
Vision For RAPP

A process of **deliberative collaboration** with a range of **diverse stakeholder groups** to:

1. Identify and test options and pathways for housing
2. Build consensus on solutions that contribute to more climate adapted, just, and disaster resilient futures

Key Objectives

- Build and demonstrate a methodology for assessing current and future multi-hazard disaster opportunities and risks
- Inform the identification of options and adaptation pathways for preventing and reducing disaster risks
- Gather findings and lessons from stakeholders to inform the recommendations to State and Local Governments and develop risk-informed Regional Planning guidelines
- Build capability and capacity with relevant stakeholders to lead and engage in deliberative adaptation pathways processes
- Shape the consideration of disaster risk reduction in strategic land use planning in NSW



Source (centre graphic): Simpson, N.P., et al., A framework for complex climate change risk assessment. One Earth, 2021. 4(4): p. 489-501.
Source (surrounding graphic): CSIRO, NSW Department of Planning, Housing and Infrastructure, Value Advisory Partners 2025

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Development of Plausible Divergent Futures

Development of Plausible Futures

- Setting the drivers and parameters of change to **make uncertainty manageable**
  - **Population** and demographics
  - **Spatial distribution** and land use
  - **Economic development**
  - **Technological change** and **energy transition**
  - Physical **Climate change**
- 'High' and 'Low' housing growth PDFs set the boundary conditions against which the provision of safe and affordable housing in the region could be considered.
- Include a range of diverse stakeholder groups to identify and test options including First Nations, community and government (both State and Local)

	Plausible Divergent Futures	Total population in 2060 - NSW	Implied Dwellings in 2060 - NSW	Net Overseas Migration - NSW	Spatial reorganisation
Low Growth	**Tales of the City**	9.9 M	4.5 M	2.3 M	More concentrated
	Survivor	10.5 M	4.8 M	2.7 M	Similar to current patterns
	Sea Change	11.5 M	5.2 M	3.1 M	More distributed
High Growth	**Better Homes and Gardens**	20.1 M	7.7 M	9.6 M	Similar to current patterns

Source: NSW2060 ITLU scenarios, NSW Department of Premier and Cabinet, 2022.

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Application of Plausible Divergent Futures Accessible Through Data Visualisation

Development of Plausible Futures

- **Creation of visualisations** representing long term housing growth within Illawarra Shoalhaven
- **Accessible and usable data** that can be provided to both Local Government and State Planning agencies
- **Localised Place-based** potential outcomes for housing within the region
- **Identifying areas of high growth** to help determine infrastructure needs through clear communication and input into prioritisation of investment

Current Dwellings
Example of developable areas in Wollongong and Shellharbour

Source: Value Advisory Partners 2023 (using data from NSW Department of Planning & Environment and Transport for NSW)

Example: Low Growth of Dwellings in 2061

Example: High Growth of Dwellings in 2061

Note: Heights of columns indicate comparative density between suburbs/towns

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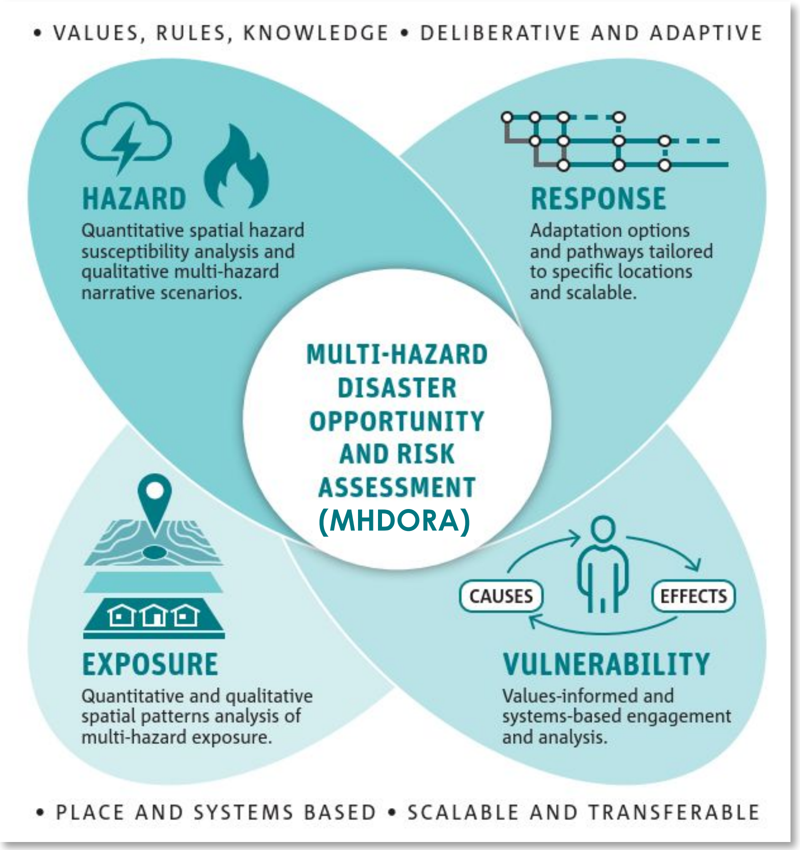
Development of a Localised Multi-Hazard Risk Assessment

Localised Multi-Hazard Risk Assessment

- **Integration of data in central hazard database** and usable for decision making
- **Capture perspectives** on the consequences of the '**things of value' being impacted.**
- **Identify factors** causing the 'things of value' to be exposed or **vulnerable to natural hazard impacts.**
- To begin to identify the options available to **reduce the future impact of natural hazards** on homes and lives in the Illawarra Shoalhaven.
- Addresses key questions relating to **climate change and regions:**
    1. Will climate change impact on current experienced hazards across the region equally?
    2. Will climate change expand the areas within the region impacted by current hazards?
    3. Will climate change create/introduce new hazards for the region it has not experienced before?

MHDORA Assessment Framework for multiple hazards within Illawarra Shoalhaven Region



Source: Wise, R.M., et al (2024) Strategically responding to climate hazards in the Illawarra Shoalhaven region, NSW. Demonstrating a multi-hazard disaster opportunity and risk assessment methodology and regional adaptation pathways planning approach in strategic land-use planning. Report prepared for the NSW Department of Planning, Housing and Infrastructure. CSIRO Environment, Australia.

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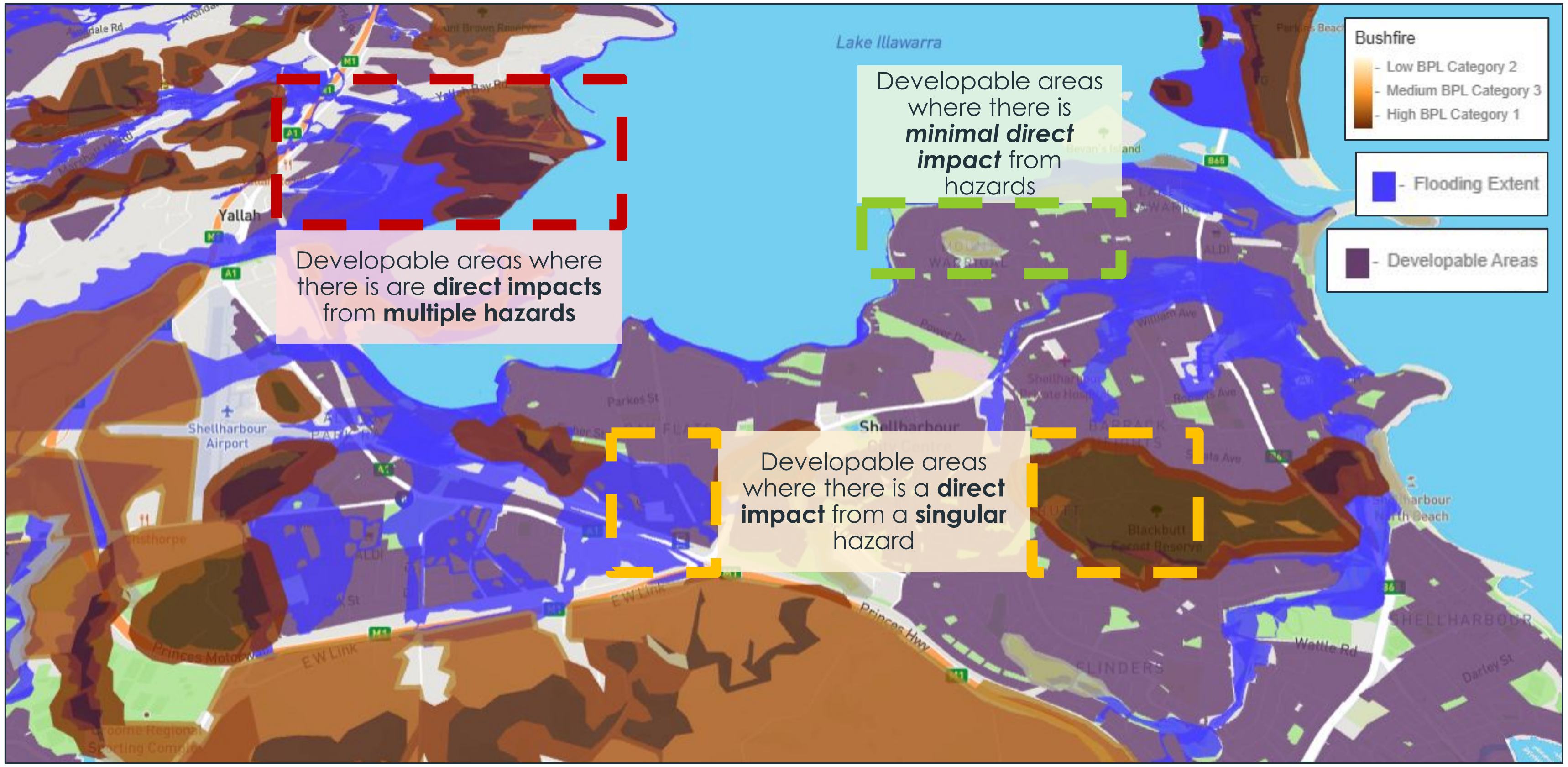
Application of Localised Multi-Hazard Risk Assessment through an Accessible Data Visualisation

Localised Multi-Hazard Risk Assessment

- **Multiple hazard data** and assessment overlays on all areas of the region
- **Accessible and usable data** that can be provided to government, emergency management and development agencies
- **Localised place-based** hazard identification to help determine long term actions on housing growth and potential risks
- **Intersection of developable areas and hazards** to help determine potential resilience infrastructure and governance needs e.g. stormwater management, building standards for developments

Example of Multiple Hazards Assessment within Shellharbour and Lake Illawarra area



Source: Value Advisory Partners 2023 (using data from CSIRO, NSW Department of Planning, Housing and Infrastructure, NSW Rural Fire Service, University of Wollongong)

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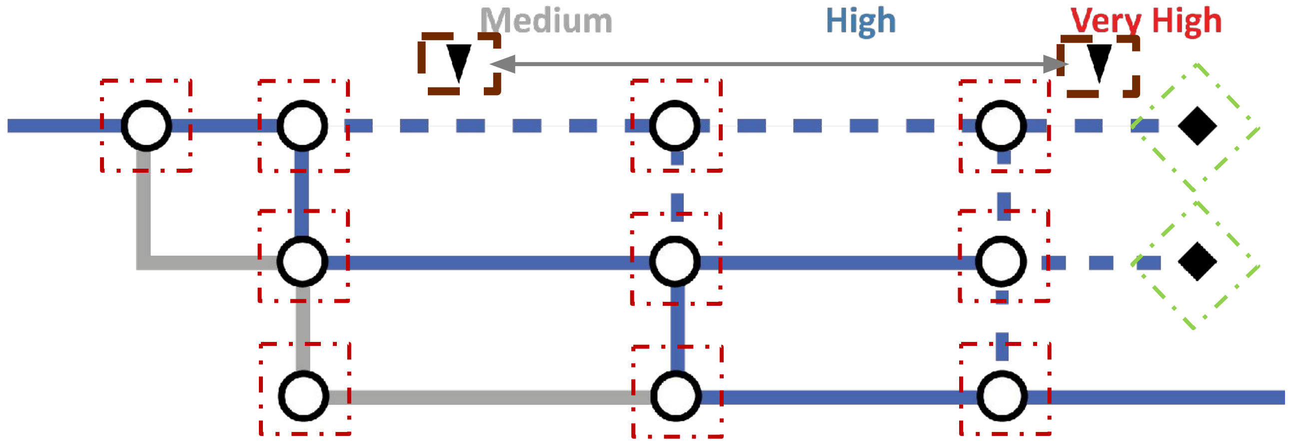
Development of Adaptation Pathways

Adaptive Pathways

DEGREE OF HAZARD



OPTION	Example of Planning Interventions
Increase Dwelling Density	Adjusted Development Overlays to increase potential Plot Area Ratios allowed
Maintain Existing Dwelling Density	No up-zoning; down-zoning; inter-council development distribution
Relocate - Buyback, Prevent Rebuilding	Setting up buy-back fund; inter-council development relocation

LEGEND

- Viable optimal housing option
- Preparatory activity
- Threshold
- Hazard Knowledge Gap
- Sub – optimal Housing
- Decision point
- Trigger

Decision points:
Indicate points at which the decision process needs to be enacted to re-evaluate the suitability of an option

Triggers:
In the form of varying degrees of hazard are used as a diagnostic device to characterise the current understanding of future hazards against the probable hazard level under a changing climate

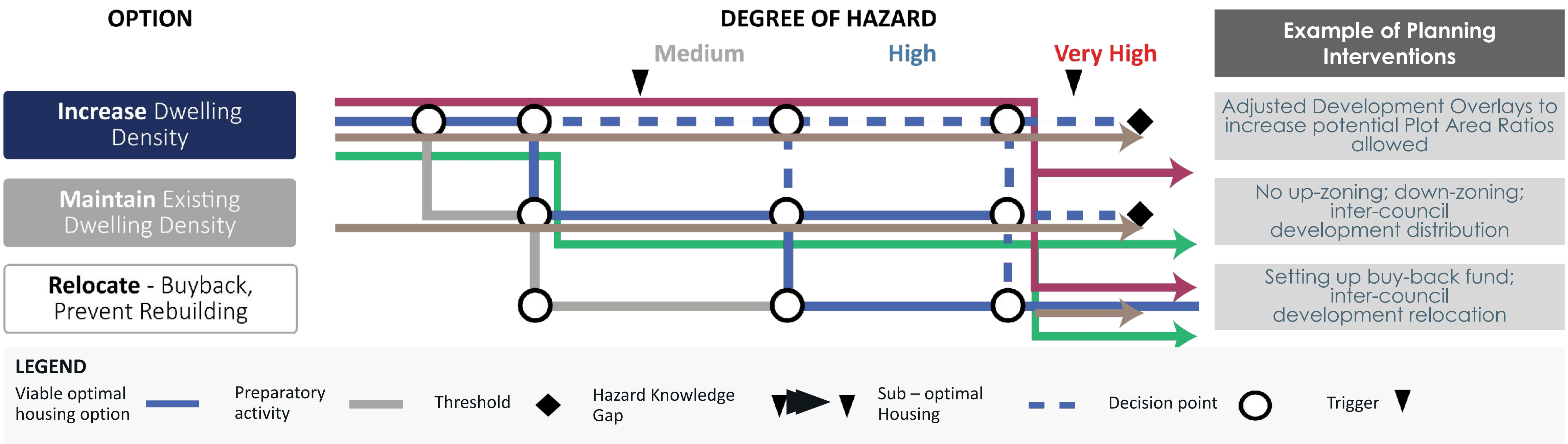
Thresholds:
Indicate the points at which the suitability of an option becomes inadequate and alternative options need to be implemented

Source; CSIRO, NSW Department of Planning, Housing and Infrastructure, Value Advisory Partners 2024

Development of Adaptation Pathways

Adaptive Pathways

The choice of pathway depends on the decision maker's level of risk-tolerance to hazard impacts



Current Policy Pathway: Low consideration and awareness of future hazard

High Hazard Tolerance Pathway: Assumes hazard forecasting is robust enough to enable effective risk mitigation

Low Hazard Tolerance Pathway: Informed by improved hazard knowledge, but focuses housing in areas with low hazard susceptibility

Source; CSIRO, NSW Department of Planning, Housing and Infrastructure, Value Advisory Partners 2024

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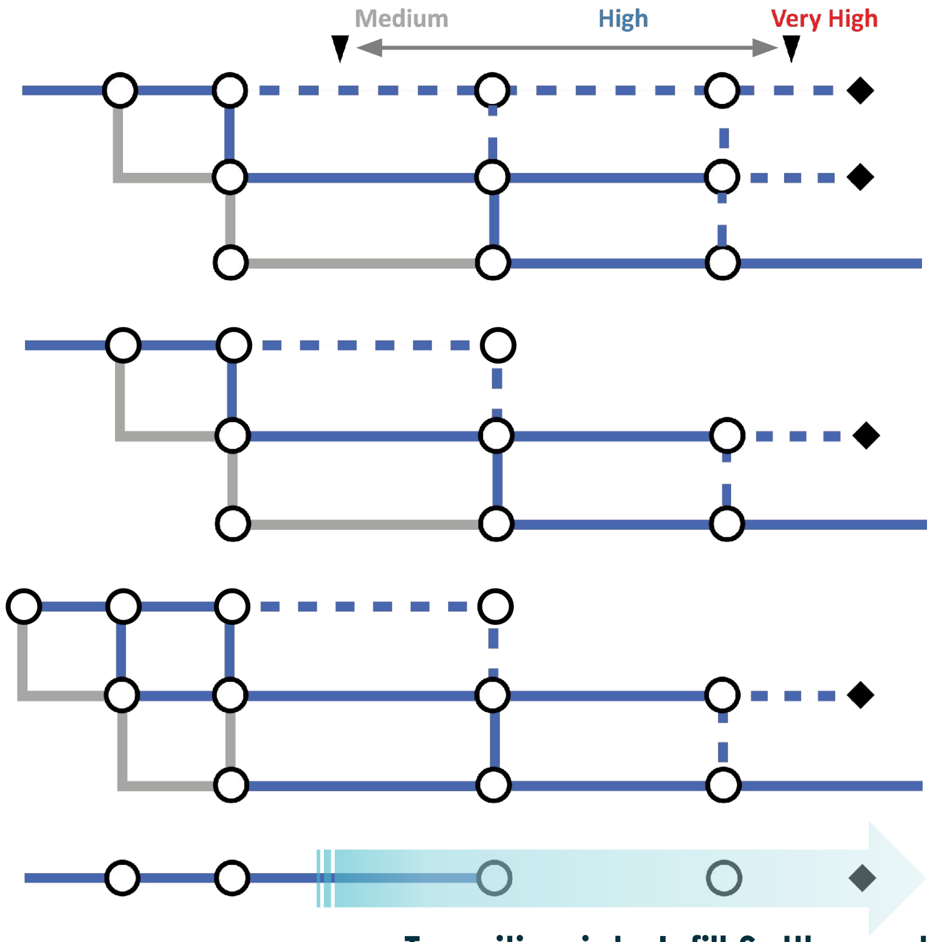
Application of Adaptation Pathways for Development in Multiple Settlement Types

Adaptive Pathways

- Each settlement type has different adaptation pathways that could be followed based on the level of risk tolerance
- Evaluation of the preferred pathways by stakeholders was based on six SDG themes
- First Nations Collective oscillated between low and high-risk tolerance
- Community Panel preferred the low risk tolerance pathway
- Government, Academics, Utilities, Finance, and Insurance participants were split between the high and low risk tolerance depending on the SDG theme

SETTLEMENT TYPES	OPTION
Infill within walkable catchments: *e,g.* Wollongong CBD	**Increase** Dwelling Density
	Maintain Existing Dwelling Density
	Relocate - Buyback, Prevent Rebuilding
Infill without walkable catchments: e.g. Balgownie (Wollongong Suburb)	**Increase** Dwelling Density
	Maintain Existing Dwelling Density
	Relocate - Buyback, Prevent Rebuilding
Expanding Townships: e.g. Mollymook	**Increase** Dwelling Density
	Maintain Existing Dwelling Density
	Relocate - Buyback, Prevent Rebuilding
Greenfield: e.g. Dapto Growth Area	**Increase** Dwelling Density

DEGREE OF HAZARD



Transition into Infill Settlement

LEGEND

Viable optimal housing option | Preparatory activity | Threshold | Hazard Knowledge Gap | Sub – optimal Housing | Decision point | Trigger

Source: NSW Department of Planning and Environment 2023; Value Advisory Partners 2025

Outcomes from RAPP and Next Steps and Opportunities for Strategic Planning

Outcomes from the RAPP Project

Key Challenges	Transformative approaches for RAPP	Strategic Planning Outcomes
• Accelerating Housing growth including regions • Increased infrastructure and amenity needs	**Development of Plausible Futures**	• Clarity on long-term population growth against capacity at a local scale • Detailed knowledge of local population needs • Determining infrastructure and amenity investment
• Increased Frequency and Intensity of Multiple Hazards • Overwhelming Effect on Regions' capabilities	**Localised Multi-Hazard Risk Assessment**	• Planning decisions reflecting short- and long-term climate hazard risks • Understand localised options to either grow, defend or retreat
• Need for new processes to deal with increased uncertainty • Planning to future proof the regions	**Development of Adaptive Pathways**	• Common framework for decision making ensuring adaptable approach to uncertain future for the regions • Being adaptable to changing policies and hazard risks when required (minimising maladaptation)

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Next Steps and Opportunities for Planning using recommendations and outcomes from RAPP Project

A New Model for Strategic planning in Urban and Regional Areas

RAPP as a pathfinder project shows how to implement similar transformative planning processes by conducting risk-informed, integrated strategic assessments in the preparation of long-term planning into other areas within Australia

Potential Opportunities: Greater Adelaide Regional Plan, Six Cities Regions Plans (NSW)

Inclusion within Disaster Adaptation Plans

RAPP's processes could be included within Disaster Adaptation Plans connecting strategic planning policies to resilience and adaptation planning, to improve housing supply in well located and protected areas

Potential opportunities : Strategies for NSW Reconstruction Authority, Emergency Management Victoria

Open Data Sources increasing accessibility of insights from processes used in RAPP

Using existing open data sources managed by governments to allow access of key hazards and planning insights from processes used in RAPP, to allow ongoing engagement with stakeholders including local community and industry

Potential opportunities : VicPlan, NSW Planning Portal, PlanSA

Collaboration with Private Industry

Assessments, outcomes and insights could be shared with private industry to ensure a coherent approach to housing development implementation which would help plan and fund infrastructure for increasing local resilience

Potential opportunities : Developers (both Greenfield and Infill), Insurance Companies

Transforming Processes Within Government

Expanding transformative processes within RAPP to other parts of government to increase capability and knowledge in planning for ongoing and uncertain changes in natural hazards, and establishing ongoing cross-sector governance for strategic assessments

Potential opportunities : Commonwealth and State Government Departments for Employment and Industry, Land Use Management, Transport and Economic Development

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Contact VAP

https://www.valueadvisory.co/
Ph: (03) 9498 0624

Peter Heinmiller
Principal
E: peter.heinmiller@valueadvisory.co

Christopher Marinopoulos
Senior Consultant
E: christopher.marinopoulos@valueadvisory.co

Designed by:
Kiara Cristiano
Consultant